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AMR-GNN: A multi-representation graph neural network framework to enable genomic antimicrobial resistance prediction

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ABSTRACT

Whole-genome sequencing (WGS) data are an invaluable resource for understanding antimicrobial resistance (AMR) mechanisms. However, WGS data are high-dimensional and the lack of standardised genomic representations is a key barrier to AMR phenotype prediction. To fully explore these high-resolution data, we propose AMR-GNN, a graph deep learning-based framework that integrates multiple genomic representations with graph neural networks (GNN) to enable AMR phenotype prediction from genomic sequence data. We test AMR-GNN with *Pseudomonas aeruginosa*, a clinically relevant Gram-negative bacterial pathogen known for its complex AMR mechanisms. We present AMR-GNN as a proof-of-concept framework designed to address several key problems in AMR phenotype prediction with data-driven machine learning (ML) approaches, including using multiple genomic representations to enhance performance, to mitigate the influence of clonal relationships and to identify informative biomarkers to provide explainability. Follow-up validation on the largest publicly available dataset spanning both Gram-negative and Gram-positive pathogens highlights AMR-GNN's broad applicability in detecting AMR in diverse and clinically relevant pathogen-drug combinations.

INTRODUCTION

Antimicrobial resistance (AMR) is a significant global health threat, leading to 1.14 million deaths globally in 2021 [1]. Rapid and accurate detection of AMR phenotype through antimicrobial susceptibility testing (AST) is crucial for improving patient outcomes, reducing unnecessary antimicrobial use, and preventing the spread of AMR in hospital settings. Culture-based methods for identifying AMR are slow, laborious and are often not reported to the clinician for 2-5 days after sample collection [2]. These delays also impact our ability to institute infection prevention measures to stop further AMR transmission.

Whole genome sequencing (WGS) data offer a promising alternative approach to detect AMR. The small size of bacterial genomes makes WGS increasingly accessible to clinical microbiology laboratories. Coupled with advances in bioinformatics, WGS data can now support a range of analyses, including organism identification and AMR detection [3]. Additionally, the constant generation of public WGS data provides opportunities for surveillance and enhances the robustness of bacterial studies, particularly when statistical analyses often require large sample sizes.

For AMR detection, most pathogens have been analysed using simple rule-based and/or gene detection approaches [4, 5]. These approaches perform well when AMR is driven by single gene presence or mutations but are less effective in pathogens where AMR results from complex interactions between several factors, or when genetic heterogeneity is present (e.g. recombination or horizontal gene transfer) [3]. They also cannot discover novel genetic variants related to AMR, potentially underutilising the high resolution of

WGS data. In recent years, machine learning (ML) has become a promising way to overcome these limitations and has successfully predicted AMR in a wide range of bacterial pathogens [6-9]. ML can account for complex genetic relationships, and once trained, can detect AMR for a broad set of antimicrobials within minutes.

Given that bacterial genomes consist of sequences containing millions of nucleotides (e.g., approximately 6 Mb for *P. aeruginosa*), WGS data can be represented in multiple ways for input into ML models. However, there is currently no established consensus on the most effective representation [4]. Using several approaches to represent bacterial genomes provides more complete insights into the pathogen. For example, single nucleotide polymorphisms (SNPs) help identify variants relative to a specific reference genome, while pan-genome analysis can uncover variant genes within a population. Despite these potential advantages, the use of multiple genome representations for AMR phenotype prediction has remained limited. Recent computational advances (e.g. deep learning) have inspired growing interest in multimodal learning [7, 10, 11], prompting the question of whether this approach can effectively exploit diverse genomic inputs. We therefore aimed to assess the impact of integrating multiple genome representations on AMR phenotype prediction, and to leverage modern ML approaches designed for complex, high dimensional data to further enhance prediction performance.

To address these gaps, we identified graph neural networks (GNNs) as a promising solution. Their flexible architecture allows the integration of diverse input types through node and edge features [12]. In contrast to other deep learning architectures, GNNs

aggregate information between samples through a similarity network. This characteristic is particularly relevant for AMR phenotype prediction as isolates sharing similar genomic features may share similar AST profiles [13]. While GNNs are increasingly applied in biomedical research [12, 14], their use in bacterial genomics for AMR phenotype prediction remains limited [15]. We therefore developed AMR-GNN, a new GNN-based framework using multiple WGS-derived features to predict AMR phenotype in *P. aeruginosa* as a use case (**Fig. 1a**). This pathogen has a large and diverse genome with high genetic variability and employs a wide range of resistance mechanisms, including efflux pumps, porin loss, and enzymatic inactivation of antibiotics [16, 17]. AMR-GNN outperformed baseline models, including elastic net and convolutional neural network (CNN), across a wide range of antipseudomonal agents. We further evaluated the model on a large, diverse dataset encompassing key Gram-negative and Gram-positive pathogens, confirming AMR-GNN's effectiveness across multiple species and antimicrobials.

Collectively, these results demonstrate the potential of leveraging distinct feature types to enhance AMR phenotype prediction, establishing AMR-GNN as a robust proof-of-concept framework for integrating these features into a unified ML architecture. Finally, through feature attribution analysis, we highlighted the interpretability of AMR-GNN in identifying AMR biomarkers.

RESULTS

Data characteristics. We collected a comprehensive dataset consisting of WGS (Illumina) and AST data from our center and nine publicly available datasets [11, 18-25]. We included datasets with minimum inhibitory concentration (MIC) values obtained from dilution protocols, either broth microdilution (BMD) (from our centre and most public datasets [18-25]) or agar dilution [11]. This resulted in a complete dataset of 2,515 *P. aeruginosa* isolates with diverse population structure (524 unique STs – **Fig. 1b**), with ST235 (10.3%), ST174 (5.5%), ST111 (4.3%) and ST244 (4.0%) being the most prevalent. The AST dataset consisted of 19,865 MIC entries covering 12 common antipseudomonal antimicrobials. These values were classified as either susceptible or resistant based on the European Committee on Antimicrobial Susceptibility Testing (EUCAST) v15.0 breakpoints [26]. The number of MIC values and resistance rates for each antimicrobial are shown in **Fig. 1c-e**. We used seven datasets (our centre and [11, 18, 20-22, 24, 25]) for training and internal validation. The two smallest datasets [19, 23] were reserved as hold-out set to assess the model's generalisability (see Methods). Resistance rates varied from 13.7% (amikacin) to 63.3% (ciprofloxacin) in the internal dataset (Fig. 1d), and from 11.1% (amikacin) to 79.4% (tobramycin) in the hold-out datasets (**Fig. 1e**).

AMR-GNN framework design. AMR-GNN predicts AMR for each isolate by treating each isolate as a node within a graph and performing node classification. Constructing this graph requires two key components: node features represented as feature vectors for each isolate and edge information captured in an adjacency matrix indicating relationships between isolates. We first evaluated AMR phenotype prediction using individual genomic representations and selected the best-performing as the node feature

input. Following a GNN-based multi-omics integration approach [27], we used the remaining representations to define graph connectivity via adjacency matrices (see **Methods**). Each graph shared the same node features but differed in connectivity, allowing us to learn distinct representations. To integrate information from both graphs, we employed low-rank multimodal fusion to combine their embedding layers into a unified representation used for final AMR phenotype prediction [28].

Models trained with a single genomic representation demonstrated a strong baseline for AMR phenotype prediction. We evaluated models trained with different single feature types, including unitigs, SNPs, and frequency chaos game representation (FCGR). FCGR generates a $2^k \times 2^k$ grid where k-mers are mapped according to their nucleotide sequence and weighted by their occurrence frequency [29, 30]. In this representation, each grid cell is analogous to a pixel in an image. For unitigs and SNPs, we applied the elastic net, while FCGR features were used to train a CNN model (see **Methods**). The complete feature sets for unitigs and SNPs contained approximately 1.5 million and 900,000 features, respectively, which were used as input into the elastic net. For FCGR, we tested k-mer values ranging 5-8 to generate FCGR features and noted that k=7 was optimal, with equivalent performance to k=8 but faster training time (**Supp. Fig. 1**).

Models using unitig features achieved the highest area under receiver operating characteristic curve (AUROC) values across all tested antimicrobials in the internal test sets, showing statistically significant improvements over other feature types for 10/12

antimicrobials, excluding cefepime and ceftolozane/tazobactam (**Fig. 2a, Supp. Table 1**). Unitig-based models exhibited strong predictive performance, particularly for fluoroquinolones (ciprofloxacin AUROC 0.911 [CI: 0.904-0.918]) and aminoglycosides (tobramycin AUROC 0.933 [CI: 0.919-0.947]). Performance for beta-lactams ranged from AUROC 0.878 (CI: 0.847-0.910) for ceftolozane/tazobactam to AUROC 0.636 (CI: 0.603-0.668) for cefepime. When comparing SNP and FCGR representations, FCGR-based models significantly outperformed SNP-based models for ciprofloxacin (0.868 vs 0.733, $P<0.001$), levofloxacin (0.875 vs 0.684, $P<0.001$), and ceftazidime (0.739 vs 0.691, $P=0.003$), with comparable performance across the remaining nine antimicrobials (**Supp. Table 2**). We then used the salient unitigs selected by the elastic net as node features, while SNP and FCGR data were used to construct the adjacency matrices for AMR-GNN (see Methods).

AMR-GNN outperformed baseline models. To evaluate the effectiveness of AMR-GNN in improving AMR phenotype prediction in *P. aeruginosa*, we used AUROC as the main metric and compared AMR-GNN with unitig-only models as the baseline (**Fig. 2b, Supp. Table 3**). AMR-GNN significantly outperformed unitig-only models in 11/12 tested antimicrobials and performed equivalently for levofloxacin (0.921 vs 0.897, $P=0.104$) (**Supp. Table 4**). Highest AMR-GNN performance was noted for tobramycin (0.971, CI: 0.964-0.978), while lowest was for cefepime (0.819, CI: 0.783-0.855). AMR-GNN made the most substantial improvements for the lowest performing antimicrobials in unitig-only models (cefepime and aztreonam), achieving AUROC increases of 28.8% and 18.9%, respectively.

Further improving performance by removing bias of bacterial population structure.

Bacterial population structure is a major confounder in AMR phenotype prediction because of slow linkage disequilibrium decay [31]. This results in co-inheritance of causal and non-causal variants that ML models cannot easily disentangle. As a result, models may rely on clonal signatures (e.g., high-risk ST clone) to predict AMR instead of uncovering the underlying resistance mechanisms. In the context of GNN, isolates within the same MLST group form dense connections due to their high genetic similarity. This can result in redundant graph information that can degrade performance [32]. Previous work has demonstrated that reducing the influence of redundant neighbors such as edge dropping or node deletion yields more robust results [32, 33]. In this study, we employed a decoupling approach that removes all edges connecting samples from the same MLST group, therefore forcing each node to aggregate information only from neighbors that are not clonally related (see **Methods**). By using this approach, we achieved higher AUROC for all tested antimicrobials compared to the original AMR-GNN, with significant improvements for meropenem, amikacin, and levofloxacin (**Fig. 2b, Supp. Table 5**).

AMR-GNN demonstrates better generalisability on external validation than unitig-

only model. After identifying decoupled AMR-GNN as our best-performing approach, we evaluated its performance and benchmarked against unitig-only models (our prior baseline comparator) on the hold-out test set for 8 antimicrobials. Compared to the internal test set, AMR-GNN maintained comparable AUROC for ciprofloxacin and ceftazidime, but showed significant performance drops for other antimicrobials (**Supp.**

Table 6), suggesting potential overfitting. However, AMR-GNN demonstrated greater robustness to overfitting than unitig-only models, outperforming them in 5/8 antimicrobials tested (**Supp. Table 7**).

AMR-GNN achieved superior performance compared with *P. aeruginosa*-specific AMR phenotype prediction approaches. We benchmarked AMR-GNN against several established approaches (see **Methods**). Rule-based methods are the current biological standard for detecting AMR based on known AMR determinants (including resistance genes and AMR causal variants) [4, 34]. We evaluated AMR-GNN against two specific implementations: the method by Cortes-Lara et al. [21], (hereafter ‘rule-based’), which utilises the presence/absence of known AMR determinants; and ARDaP [35], an automated scoring system combining known determinants with variants identified via genome-wide association studies (GWAS) and ML-based feature selection. Notably, ARDaP previously outperformed other common AMR phenotype prediction tools (e.g., abritAMR, AMRFinderPlus, and ResFinder) in all 10 tested antimicrobials. In addition, although we previously demonstrated AMR-GNN’s superior performance over elastic net and CNN models, we also compared it with VAMPr, a publicly available AMR phenotype prediction tool that uses KEGG (Kyoto Encyclopedia of Genes and Genomes) ortholog variants as features [36].

Based on label availability, we benchmarked AMR-GNN against a rule-based approach, ARDaP, and VAMPr using both internal and external test sets (**Fig. 3**). On the internal test set, AMR-GNN achieved universally higher F1-scores and better very major error

rates (VME) across all tested antimicrobials (n=5, 8, and 8, respectively). In the external validation (comprising 5, 6, and 6 antimicrobials), AMR-GNN demonstrated superior F1-scores in 4/5, 3/6, and 4/6 cases, respectively. For VME, AMR-GNN achieved better results for 5/5, 4/6, and 4/6 cases, respectively. AMR-GNN underperformed other approaches for meropenem resistance prediction (F1: 0.562, CI: 0.478-0.647). We noted meropenem resistance was significantly more prevalent in the hold-out data than in the internal set (54.0% vs 31.4%; $P < 0.001$, Chi-square test), which likely contributed to this performance drop.

AMR-GNN demonstrated robust predictive performance across several key bacterial pathogens. To assess the efficacy of AMR beyond *P. aeruginosa*, we evaluated its performance using extensive datasets from two Gram-negative organisms [*Escherichia coli* [n=10,246] and *Klebsiella pneumoniae* [n=7,072]] and two Gram-positive organisms [*Staphylococcus aureus* [n=3,195] and *Enterococcus faecium* [n=2,608]]. The analysis covered a comprehensive panel of antimicrobials specific to each species, sourced from The Bacterial and Viral Bioinformatics Resource Center (BV-BRC) databases [37, 38] (**Fig. 4a-d**; see **Methods**). AMR-GNN consistently achieved high predictive performance, with mean AUROCs exceeding 0.9 for nearly all tested antimicrobials, except for piperacillin/tazobactam in *E. coli* (0.871, CI: 0.861-0.881) (**Fig. 4e-h**). For each species, highest AUROCs were noted for ceftriaxone in both *E. coli* (0.988; CI: 0.986-0.989) and *K. pneumoniae* (0.985, CI: 0.981-0.990), fusidic acid in *S. aureus* (0.991, CI: 0.988-0.994); and vancomycin in *E. faecium* (0.995, CI: 0.993-0.997). Clinical decision metrics were similarly strong: sensitivity/specificity of 0.938/0.968 for *E. coli*-

ceftriaxone, 0.853/0.986 for *K. pneumoniae*-ceftriaxone, 0.990/0.909 for *S. aureus*-fusidic acid, and 0.934/0.992 for *E. faecium*-vancomycin (**Supp. Table 8**). VME values ranged from 0.008 to 0.133, while ME values ranged from 0.054 to 0.336. In terms of benchmarking, we compared the performance of AMR-GNN with ResFinder, a tool that has shown strong results relative to several other AMR phenotype prediction methods [5]. Specifically, we assessed predictions across 16 species-antimicrobial combinations. In terms of F1-score, AMR-GNN outperformed ResFinder in 14 of these 16 comparisons. The only exceptions were tobramycin in *E. coli* (0.834 vs. 0.859) and fusidic acid in *S. aureus* (0.929 vs. 0.969) (**Fig. 4i**). The most significant performance improvement with AMR-GNN was observed for linezolid in *E. faecium* (0.859 vs 0.512).

Model interpretation on best-performing antimicrobials. We employed integrated gradient (IG) analysis to identify the genetic features contributing most to AMR-GNN performance [39]. We selected the highest-performing antimicrobials from three key antipseudomonal classes: fluoroquinolones (levofloxacin), aminoglycosides (tobramycin), and beta-lactams (ceftolozane/tazobactam) (**Table 1**) [39]. For levofloxacin, the model *a priori* identified known target sites (*gyrA*, *gyrB*, and *parC*) as the most important contributors [40], while PA4179 encodes a probable OpdK porin allowing antimicrobial transport across the bacterial membrane [41, 42]. For tobramycin and ceftolozane/tazobactam, most shortlisted genes had undetermined functions. *fusA1* (tobramycin) has been shown to lead to downstream modification that result in a 4- to 8-fold increase in the MIC for several aminoglycosides, including tobramycin [43]. For ceftolozane/tazobactam, PA4520 is located immediately upstream of *ampE* and *ampD*,

which may allow over-expression of AmpC beta-lactamases. We further validated model interpretation results by assessing impact of mutations in shortlisted genes of known functions. Compared to wild-type isolates, MICs were significantly higher in isolates with mutations: *gyrA* ($P=2.655 \times 10^{-134}$), *gyrB* ($P=6.959 \times 10^{-5}$), *parC* ($P=1.053 \times 10^{-103}$) in levofloxacin, and *fusA1* ($P=1.452 \times 10^{-4}$) in tobramycin (Mann-Whitney U-test) (**Supp. Fig. 2a,b**). In terms of specific mutations, we identified 9, 2, and 4 mutations in *gyrA*, *gyrB*, and *parC*, respectively, that were significantly enriched in levofloxacin-resistant isolates (Chi-square test; **Supp. Table 9**). Additionally, 7 mutations in *fusA1* were significantly associated with tobramycin resistance (**Supp. Table 9**).

DISCUSSION

While ML models leveraging WGS data show promise for AMR phenotype prediction, most existing approaches rely on single genomic representations, underutilising the data's high-dimensional potential. To address this, we developed AMR-GNN, a unified framework integrating multiple genomic feature sets. This multi-representation strategy significantly outperforms conventional and rule-based methods in *P. aeruginosa*, as validated by our benchmarking studies. AMR-GNN leverages two key properties of GNNs: their ability to simultaneously optimise classification outcomes while learning meaningful network structure [14], and their flexibility in processing diverse input data types [12]. This enables our model to not only generate informative AMR embeddings but also capture shared genetic content among bacterial isolates. Moreover, AMR-GNN mitigated population structure bias through intra-ST edge removal. Finally, the framework

demonstrates species-agnostic transferability, accurately predicting resistance patterns across diverse bacterial pathogens.

We demonstrated that AMR-GNN successfully predicts AMR for *P. aeruginosa*, a key nosocomial pathogen with complex mechanisms conferring AMR that have challenged previous AMR phenotype prediction efforts. Prior ML efforts were constrained by small cohorts [36] or limited drug coverage [11, 36, 44, 45]. To circumvent these gaps, we curated a large, geographically diverse set of *P. aeruginosa* genomes paired with gold-standard AST results and trained AMR-GNN using multiple, complementary genomic representations. Consistent with previous work [7], we found that reference-free genomic features (i.e., unitigs) achieved the best performance among single-feature models. However, we also observed that incorporating pairwise genomic distances, which are widely used for detecting outbreaks [46], as edge features significantly improved predictive power. This multi-view integration likely accounts for the model's superior performance over single-feature models and *P. aeruginosa*-specific rule-based approaches across all tested antipseudomonal agents. These findings encourage the integration of additional data types to further enhance performance. For instance, although it is costly to obtain, incorporating transcriptomic data as node features warrants consideration for *P. aeruginosa* given its capacity for environment-driven transcriptional adaptation [47].

The observed performance drop in the external test set indicates that larger training datasets are likely required to improve generalisation, particularly for species with

complex AMR mechanisms like *P. aeruginosa*. Resistance mechanisms display marked regional specificity, meaning that variants widespread in one location may be absent elsewhere [48]. This likely explains why ARDaP, which uses a comprehensive, globally curated AMR determinant database, outperformed AMR-GNN in 3 out of 6 external comparisons. These results therefore highlight the necessity of continuous data integration and model updating to ensure robustness across different geographical regions and time periods.

Beyond *P. aeruginosa*, AMR-GNN's performance across more than 23,000 genomes spanning four clinically important pathogens, tested against a comprehensive panel of species-specific antimicrobials, underscores the framework's breadth and adaptability. Mean AUROC values surpassed 0.90 for almost every species-drug pair, despite the biological and genomic differences between *E. coli*, *K. pneumoniae*, *S. aureus* and *E. faecium*. This indicates that the graph-based integration of multiple genomic representations, a core design choice of AMR-GNN, successfully captures diverse resistance mechanisms of both Gram-negative bacilli and Gram-positive cocci. Furthermore, we noted the top-scoring combinations (i.e., ceftriaxone in *E. coli*, ciprofloxacin in *K. pneumoniae*, and vancomycin in *E. faecium*) are of significant clinical interest. Although VME and ME are critical for clinical translation, they are rarely detailed in AMR phenotype prediction literature. Compared to reported values for *K. pneumoniae* and *S. aureus* [49], our model achieved better VME rates in most cases (5/6 and 3/3, respectively). The mixed performance regarding ME (superior in 2/6 and 1/3 comparisons)

is attributable to our threshold optimisation process, which assigned a higher penalty weight to VME.

Regarding model interpretation, we assessed IG attribution scores to identify salient biomarkers for antimicrobials where AMR-GNN achieved high performance in *P. aeruginosa*. While key AMR drivers were identified in models predicting resistance to levofloxacin and tobramycin, the analysis of ceftolozane/tazobactam resistance revealed several genes of unknown function, emphasizing the complexity of beta-lactam resistance where resistance likely emerges from the interaction of multiple mutations [50]. It might also explain the superiority of AMR-GNN over rule-based approaches in predicting ceftolozane/tazobactam resistance, as the latter relies on known AMR variants. By deriving unitig-based node features without requiring *a priori* AMR knowledge, AMR-GNN eliminates the need for extensive curation of resistance determinants and may facilitate the use of ML approaches for AMR surveillance. Furthermore, the observation of significantly higher MICs in isolates with mutations in the identified AMR determinants further validates the model's interpretability and biological relevance. These findings not only confirm the effectiveness of our AMR-GNN model in predicting AMR but also underscore its interpretability which is crucial for consideration in clinical applications.

We acknowledge several limitations of our study. First, constructing node features requires a prior feature-selection step, and these chosen features may change as new isolates are added to capture evolving AMR variant profiles. While new isolates can be queried against an existing unitig list, this approach may obscure novel biological signals.

For instance, if two isolates have distinct mutations at the same nucleotide position, the reference unitig would be recorded as “absent” for both. Without updating the database to include the specific unitigs for these new variants, the genetic nuance distinguishing these isolates is lost. Second, confirming the role of newly identified genes with unknown functions will require laboratory-based validation, which are beyond the scope of this study. Third, while our approach yielded performance improvements, we acknowledge the inherent limitations of using ST to account for population structure bias. While ST data is widely available and generally informative, its dependence on a limited set of housekeeping genes disregards significant genomic variation. Future iterations could therefore benefit from employing higher-resolution typing schemes, such as core-genome MLST (cgMLST), to better address lineage biases. Finally, a prospective study is needed to assess AMR-GNN’s utility in guiding real-world treatment decisions. Since its performance varies by species-drug combination, we therefore recommend first applying the framework to those combinations with the strongest predictive performance, such as fluoroquinolones and aminoglycosides across multiple species, before broader deployment.

In conclusion, our study demonstrates that the performance of ML-based AMR phenotype prediction models can be significantly improved through both the selection of appropriate WGS feature representations and the use of advanced deep learning architectures. Instead of relying on a single feature representation, integrating multiple genomic features offers distinct advantages. AMR-GNN presents an interpretable and flexible framework for such integration, with promising potential for application to multi-omics modalities such

as proteomics and transcriptomics, which have also demonstrated utility in predicting AMR phenotype [11, 51, 52]. Although this study focuses solely on WGS data, the modular design of AMR-GNN readily accommodates additional data sources, such as electronic health records for patient network modeling in outbreak detection.

METHODS

Data collection. This study was approved by the Alfred Hospital Ethics Committee (Project No. 185/21). We collected short-read (Illumina) genomic data and broth microdilution (BMD) results (Sensititre™ Gram-negative plate, Thermo Fisher; plate code: DKMNG) for 341 *P. aeruginosa* isolates previously obtained at Alfred Hospital between 2004 and 2021. As these data were sourced from a biorepository (Alfred Hospital Ethics Committee Project No. 533.16), patient consent was waived. Additionally, we collected data for 2,174 isolates from nine publicly available datasets (NCBI project accession numbers: PRJEB21341, PRJEB31047, PRJEB37711, PRJEB40140, PRJEB61879, PRJNA264310, PRJNA288601, PRJNA526797, and PRJNA656645) [11, 18-25]. Together, this accounts for the 2,515 *P. aeruginosa* isolates included in the analysis. We used sra-tools v3.0.0 (<https://github.com/ncbi/sra-tools>) to download the public sequencing read sets. We converted MIC values to binary label (susceptible/resistant) using the EUCAST v15.0 breakpoint tables [26].

Training, testing and hold-out dataset generation. First, for external validation, we identified two collections with the smallest sample sizes (42 and 21 isolates) and set them aside as a hold-out set (referred to as “external test set”) [19, 23]. The hold-out dataset

contained AST results for nine antimicrobials; however, we excluded imipenem from the external validation due to absolute resistance rate (100%). Consequently, we conducted external validation for eight antimicrobials: meropenem, tobramycin, amikacin, ciprofloxacin, ceftolozane/tazobactam, ceftazidime, piperacillin/tazobactam, and aztreonam. We split the remaining data into training and testing sets using an 80/20 stratified split based on AST labels for each antimicrobial. This test set is referred to as internal test set. To account for performance variability, this process was repeated 10 times with random seeds. For consistency, the same training/testing splits were used across all experiments, including models based on single genomic representations, AMR-GNN training, and benchmarking with external AMR phenotype prediction tools. For AMR-GNN and the CNN model using FCGR inputs, the training set was further divided into training (80%) and validation (20%) subsets. These models were trained on the reduced training set to minimise validation loss. The final models were evaluated on both internal and external test sets.

WGS processing. We used Unicycler v0.5.0 to assemble short-read WGS data into contigs [53]. The assemblies were then uploaded to Pathogenwatch (<https://pathogen.watch>) to filter out contaminated isolates, ensuring that only *P. aeruginosa* isolates were included in the analysis, and to obtain MLST results. For unitig input, we generated a unitig database using unitig-caller v1.2.1 [54]. To reduce memory requirements associated with large unitig sets, we filtered out unitigs that were either too rare (present in less than 5% of isolates) or too common (present in more than 95%). This filtering reduced the total number of unitigs from 5,590,988 to 1,537,357. Variant calling

was performed using Snippy v4.4.5 (<https://github.com/tseemann/snippy>) which generated a SNP profile, including single nucleotide polymorphisms and indels, for each isolate using the PAO1 reference genome (Genbank: NC_002516.1). The individual variant call format (VCF) files were then merged into a single VCF input file using BCFtools v1.20 for the prediction step with pyseer v1.3.10 [55]. FCGR was used to encode the frequency of k-mers into an image-like representation [30]. Following the approach described by Green et al. for *Mycobacterium tuberculosis* [6], we extracted sequences of 57 genes known to be associated with antimicrobial resistance (AMR) in *P. aeruginosa*. The curated gene list can be found in **Supp. Table 10** [16, 21]. The gene sequences were aligned to the PAO1 genome and extracted from the Snippy output. Each gene sequence was then converted into an FCGR matrix, and all matrices were stacked along a new axis. This process resulted in an image-like input with dimensions (C, H, W), where C represents the channel dimension (i.e., the number of AMR genes, $n=57$), and $H=W=k^2$, with k being the size of the k-mer used to extract FCGR features. We tested k values ranging from 5 to 8 and determined that 7-mer was the optimal k-mer setting.

Machine learning model development for single genomic representation. We trained models to predict AMR using single genomic representations and selected the best-performing representation as our baseline benchmark. Here, we extracted multiple WGS-derived features, including unitig, SNP and FCGR [30]. Such genomic representations have each demonstrated efficacy in AMR phenotype prediction and potentially provide different strengths in representing bacterial genomic information [9, 56, 57]. For example,

SNPs identify changes in single positions [11] but depend on comparison with a reference genome, while unitigs offer an unbiased, non-reference approach [56, 58]. On the other hand, image-like representations such as FCGR may be more suitable for deep learning models [6, 9, 57]. For unitig and SNP features, we used pyseer's implementation (v1.3.10) of the elastic net model with the flags `--wg enet` and `--alpha 1` [59]. For FCGR representation, we used the same CNN architecture as described by Ren et al. [9], consisting of 3 convolutional layers and a final classification head. To compute the loss, we used binary cross-entropy function (BCELoss) adjusted by class-specific weights. We employed mini-batch gradient descent with a batch size of 32, which is regarded as a robust empirical baseline [60, 61]. We trained our models for 200 epochs. We used the Adam optimizer with a multi-step learning rate, starting from 10^{-3} and reducing to 10^{-4} and 10^{-5} after the 50th and 100th epochs, respectively. The best performing model is the one minimizing the validation loss. We used the following packages to develop our models: PyTorch v2.2.0, PyTorch Lightning v2.1.3, scikit-learn v1.4.0, and Weights & Biases (wandb) v0.16.3.

Model evaluation and statistical analysis. We validated our models with 10-fold random splits to ensure robustness. We selected AUROC as our primary metric to assess prediction performance. If models only predict binary AMR phenotype, we evaluated F1-score. Additionally, we reported the area under the precision-recall curve (AUPRC), sensitivity, specificity, and log-loss. The classification threshold was chosen to maximize the F1-score. In addition, we also reported clinically related metrics, i.e., VME and ME:

$$\text{Very major error rate (VME)} = \frac{\text{Resistant isolates falsely predicted as susceptible}}{\text{Total number of resistant isolates}} \quad (1)$$

$$\text{Major error rate (VME)} = \frac{\text{Susceptible isolates falsely predicted as resistant}}{\text{Total number of susceptible isolates}} \quad (2)$$

The optimal threshold was determined by minimising the weighted sum of VME and ME:

$$\text{Optimal threshold} = \text{argmin}(\lambda \text{VME} + \text{ME}) \quad (3)$$

Given that the Food and Drug Administration (FDA) establishes stricter acceptable thresholds for VME (1.5%) compared to ME (3%) for new AST systems [62], we applied a weighting factor of $\lambda=2$ to prioritise VME minimisation. All metrics were reported as mean values with 95% confidence intervals (95% CI). Unless otherwise specified, we used the Mann-Whitney U test to compare performance between two different feature types or model architectures. A significance level of $P=0.05$ was applied.

AMR-GNN development. We employed graph convolutional neural network (GCN) for multi-genomic representation learning [63]. The graph network requires two input types: a feature vector for each node and an adjacency matrix that defines the bi-directional edges connecting pairs of nodes. For each antimicrobial, we constructed the node features using the unitigs selected by elastic net model. Specifically, we retained all unitigs with non-zero beta coefficients. The feature data was represented as a binary feature vector, where a value of 0 indicated the absence of the corresponding unitig in the genome, while a value of 1 denoted its presence.

For edge construction, we first created distance matrices for both SNP and FCGR representations using Hamming code and Euclidean distance, respectively. Next, we arranged the unique distances in ascending order. Considering n as the total number of samples and $k\%$ as the desired proportion of edges to be retained in the final graph, we

calculated the total number of possible undirected edges ($|E|_{\text{total}}$) and the number of edges to be kept ($|E|_{\text{kept}}$) using the following equations:

$$|E|_{\text{total}} = \frac{n \times (n-1)}{2} \quad (4)$$

$$|E|_{\text{kept}} = \text{int}\left(\frac{|E|_{\text{total}} \times k}{1000}\right) \quad (5)$$

For each distance matrix, we set the distance threshold to determine the adjacency matrix. We incrementally increased the distance threshold, connecting two nodes if their distance was less than the current threshold. At each step, we calculated the total number of edges in the resulting adjacency matrix ($|E|_A$). We continued this process until the number of edges exceeded a predefined value $|E|_A > |E|_{\text{kept}}$. The adjacency matrix obtained from the previous threshold was then selected as the final adjacency matrix. This matrix was converted into the COO (Coordinate list) format, which is compatible with pytorch-geometric's data object (v2.4.0). For the hyperparameter k , we gradually increased the threshold until $k=20$, as the performance deteriorated significantly beyond that for all antimicrobials. We found that $k=2$ provided the best results (**Supp. Fig. 3**).

In terms of model architecture, we observed that adjacency matrices derived from different genomic representations varied, particularly at lower retention thresholds (k) (**Supp. Fig. 4**), suggesting that each graph might capture distinct aspects of the data. We therefore developed a dual GCN model that integrates the representation layers from two separate GCN modules into a unified vector input (**Figure 2**). This combined vector serves as the input to predict AMR. Each GCN module learns from the unitig node features and an adjacency matrix, which is inferred from either SNP or FCGR feature.

The latent representation h_i , obtained from each GCN module, can be expressed as follows:

$$h_i = \text{GCN}(X, A_i) \quad (6)$$

where $i \in [SNP, FCGR]$, X and A_i are the node feature and adjacency matrix, respectively. Specifically, each GCN module consists of two GCN layers, each containing 128 units with rectified linear units (ReLU) activation. We applied layer normalisation and used a dropout rate of 0.5. This generated two 128-dimensional representation vectors prior to the classification head. In literature, various approaches have been explored to combine multiple representations. Among these techniques, tensor presentation has demonstrated superior performance compared to simple fusion methods such as concatenation or pooling. However, such an approach often increases the computational resource exponentially with respect to the number of data modalities. To address this challenge, we adopted the low-rank fusion technique, a tensor representation approach introduced by Liu et al. [28]. This method allows us to efficiently combine the learned representations from two graph networks into the final layer preceding the classification head while maintaining computational efficiency. We used PyTorch Geometric v2.4.0 to develop our AMR-GNN model.

Clonal relationship decoupling process. We first used MLST information to construct a MLST matrix (M) similar to an adjacency matrix, where $M_{i,j} = 1$ if the i^{th} isolate from the row and the j^{th} isolate from the column are in the same MLST group, and $M_{i,j} = 0$ otherwise. We then used the M matrix to update the adjacency matrix (A) such that $A_{i,j} = 0$ if $M_{i,j} = 1$, resulting in the final decoupled adjacency matrix.

AMR phenotype prediction with rule-based approaches. As no pipeline was provided for the approach by Cortes-Lara et al. [21], we developed our own workflow to classify isolates as susceptible or resistant to five antimicrobials: meropenem, tobramycin, ciprofloxacin, ceftazidime, and ceftolozane/tazobactam. For each antimicrobial, we evaluated the known AMR determinants and assigned scores based on a reference table. Isolates with a cumulative score of 1 or higher were categorized as resistant, while those with scores below 1 were classified as susceptible. We utilized the variant calling results obtained from Snippy to assess chromosomal mutations and queried the genes of interest using the gene annotation results from bakta to determine gene presence or absence [64]. We excluded all natural polymorphism variants, as defined by Cortes-Lara et al., when calculating the scores. Regarding ARDaP [35], we ran the Nextflow pipeline (<https://github.com/dsarov/ARDaP>) to generate the report and extracted the predicted phenotypes for the tested antimicrobials.

AMR phenotype prediction with VAMPr. We performed command-line VAMPr AMR phenotype prediction [36], using the approach described (<https://github.com/jiwoongbio/VAMPr>). AMR phenotype prediction models of amikacin, aztreonam, cefepime, ceftazidime, imipenem, piperacillin/tazobactam, and tobramycin can be found at <https://cdc.biohpc.swmed.edu/VAMPr/VAMPr.cgi>.

AMR-GNN validation on BV-BRC database. Assemblies of tested isolates were downloaded from the FTP server at <ftp://ftp.bvbrc.org> [37, 38]. For Gram-negative

pathogens (*E. coli* and *K. pneumoniae*), we evaluated susceptibility to cefepime, ceftriaxone, piperacillin/tazobactam, meropenem, and tobramycin. For *E. faecium*, vancomycin and linezolid were tested, whereas for *S. aureus*, we included oxacillin, fusidic acid, ciprofloxacin, clindamycin, daptomycin, rifampicin, and trimethoprim/sulfamethoxazole. To standardize susceptibility labels across datasets, isolates with available MIC values were reclassified according to EUCAST v15 clinical breakpoints [26]. For those having only AST labels, samples labeled as intermediate ("I") were excluded from analysis as it was unclear which interpretive criteria were used. For each organism-antimicrobial combination, we followed the same approach used previously for *P. aeruginosa*: datasets were split into training and testing subsets, and models were trained across 10 random splits. During feature (unitig) selection for *E. coli*, the training dataset was limited to 3,000 isolates due to memory constraints associated with large variant files. Reference genomes used for variant calling were: *E. coli* str. K-12 substr. MG1655 (Genbank: U00096.3), *K. pneumoniae* subsp. *pneumoniae* MGH 78578 strain ATCC (Genbank: CP000647.1), *E. faecium* strain SRR24 chromosome (Genbank: CP038996.1), and *S. aureus* subsp. *aureus* NCTC 8325 (Genbank: CP000253.1). Known resistomes for FCGR feature extraction were derived from the curated Antibiotic Resistance Ontology (ARO) gene list from the Comprehensive Antibiotic Resistance Database (CARD) (**Supp. Table 11-14**) [65].

Model interpretation.

We assessed node feature importance using IG [39]. IG adheres to two key interpretability axioms: (1) sensitivity, ensuring identical attributions for models with identical outputs;

and (2) completeness stating that the total attributions equal the output difference between the models using actual input and baseline input (e.g., a zero input). Here, we used Captum's implementation (v0.7.0) of IG to compute the attribution scores [66].

To link salient node features (unitigs) to biological meaning, we performed BLAST searches to identify corresponding genes [67]. The top 10 genes were prioritized based on the cumulative IG scores of their mapped unitigs. We then used decision plots derived from SHAP (Shapley Additive Explanation) for interpretation [68].

Data availability.

The in-house sequence data generated in this study have been deposited in the NCBI BioProject ID PRJNA1220180. The Sequence Read Archive (SRA) accession numbers and AST data of all isolates can be found in Supplementary Data 1.

Code availability

The source code of this study is available at <https://github.com/andyvng/amr-gnn>. A citable version with a DOI is available on Zenodo (<https://doi.org/10.5281/zenodo.18500983>)[69].

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Author contributions

HA.N, A.Y.P. and N.M. conceived the study. J.A.W. designed and supervised sampling and collection of bacterial isolates. J.A.W., L.V.B., G.Z.B, R.T., N.Q.D., and M.H.P. collected the bacterial isolates, performed bacterial characterization, conducted whole genome sequencing and broth microdilution. HA.N. preprocessed data. HA.N., J.S., D.L.D., A.G.G, Z.W, X.W and N.M. conceptualized machine learning analyses. HA.N. developed machine learning models and evaluated predictive performances. HA.N. and N.M. analysed all results. HA.N. and N.M. wrote the manuscript with comments and feedback of all of the co-authors. All authors read and approved the manuscript.

Competing interests

All authors declare no conflict of interest.

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Tables

Table 1. List of the top 10 contributing genes for AMR phenotype prediction in levofloxacin, tobramycin, and ceftolozane/tazobactam.

For each antimicrobial, genes are displayed in descending order of IG scores.

Antimicrobial	Gene/Locus tag	IG score	Function
Levofloxacin	<i>gyrA</i>	12.154	Fluoroquinolone target (DNA gyrase subunit A)
	<i>gyrB</i>	5.027	Fluoroquinolone target (DNA gyrase subunit B)
	<i>parC</i>	3.783	Fluoroquinolone target (topoisomerase IV subunit A)
	PA4179	3.551	Encoding OpdK porin [41, 42]
	PA4913	2.103	
	PA4698	1.754	
	PA2497	1.736	
	PA4709	1.662	
	<i>ftsY</i>	1.560	Absence in MPAO1 reference genome [70]
	PA2815	1.537	
Tobramycin	PA2704	3.100	
	PA0677	3.040	
	<i>fusA1</i>	2.888	4- to 8-fold increase in aminoglycoside MIC [43]
	PA2330	2.846	
	PA1259	2.076	
	PA2338	1.725	
	PA2040	1.574	

Ceftolozane/ tazobactam	<i>catB</i>	1.483	
	<i>gyrA</i>	1.438	Fluoroquinolone target (DNA gyrase subunit A)
	<i>opdB</i>	1.382	
	PA4520	3.292	Immediate upstream neighbor of <i>ampE</i> (PA4521) and <i>ampD</i> (PA4522)
	PA1270	3.042	
	PA1334	3.028	
	PA3733	2.970	
	<i>fruA</i>	2.726	
	PA0429	2.722	
	<i>rpoB</i>	2.717	Rifampicin target (DNA-dependent RNA polymerase β -subunit) [71]
	PA4506	2.677	
	PA1245	2.514	
	<i>ccpR</i>	2.465	

Figure Legends/Captions

Fig 1. Overview of the AMR-GNN workflow and *P. aeruginosa* dataset used for training/testing

a. Schematic representation of AMR-GNN framework. Created in BioRender. Nguyen, H. (2026) <https://BioRender.com/wugkksb>. **b.** Phylogenetic tree based on core gene alignment of all *P. aeruginosa* isolates. The inner ring indicates the dataset origin of each isolate. The outer ring represents the multi-locus sequence types (MLSTs). **c.** Number of isolates per antimicrobial in the internal and hold-out dataset. **d, e.** Distribution of susceptible and resistant labels per antimicrobial the internal (d) and hold-out (e) dataset. Abbreviations: AMI: amikacin; AMR: antimicrobial resistance; AZT: aztreonam; C/T: ceftolozane/tazobactam; CEP: cefepime; CIP: ciprofloxacin; CZA: ceftazidime/avibactam; IMP: imipenem; LEV: levofloxacin; MER: meropenem; MLST: multi-locus sequence type; P/T: piperacillin/tazobactam; TAZ: ceftazidime; TOB: tobramycin.

Fig 2. Performance evaluation of individual genomic features and AMR-GNN

a. AUROC and F1 performance using individual genomic features (see **Supp. Table 1** for the complete metrics). Bars indicate the mean performance across 10 random splits, while dots represent the results from each individual split. The best-performing single genomic representation (unitig) is selected for comparison with AMR-GNN. **b.** Performance comparison of AMR-GNN (with/without decoupling method) against baseline models (unitig-only models). Color shading denotes the performance difference between the baseline models and the decoupled AMR-GNN, which is our best performing-model. See **Supp. Table 3** for the complete metrics.

Abbreviations: AMI: amikacin; AZT: aztreonam; AUROC: area under the receiver operating characteristic curve; AUPRC: area under the precision-recall curve; CEP: cefepime; CIP: ciprofloxacin; C/T: ceftolozane/tazobactam; CZA: ceftazidime/avibactam; FCGR: frequency chaos game representation; IMP: imipenem; LEV: levofloxacin; MER: meropenem; P/T: piperacillin/tazobactam; SNP: single nucleotide polymorphism; TAZ: ceftazidime; TOB: tobramycin.

Fig 3. F1-score performance of AMR-GNN compared to rule-based approaches and publicly

available AMR phenotype prediction models. Performance on both the internal and hold-out test sets

is shown. Bars indicate the mean performance across 10 random splits, while dots represent the results from each individual split.

Fig 4. AMR-GNN performance on key bacterial pathogens from the BV-BRC database.

a-d. Phenotypic AST labels (S:susceptible/R:resistant) of BV-BRC dataset across tested antimicrobials for each organism. **e-h.** AUROC performance for each antimicrobial per organism, showing the best curve from 10 random splits. **i.** F1-score performance of AMR-GNN compared to Resfinder. Bars indicate the mean performance across 10 random splits, while dots represent the results from each individual split.

Editor's Summary

Predicting antimicrobial resistance from bacterial genomic data is challenging. Here, the authors introduce AMR-GNN, a graph neural network that integrates multiple genomic representations to improve prediction, reduce clonal bias, and identify biomarkers.

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